9. 2 Infinite Series

- Sums that involve infinitely many terms: $a_1 + a_2 + a_3 + \ldots + a_n + \ldots = \sum_{n=1}^{\infty} a_n$.

- Unlike finite sum, NOT ALL infinite series have a sum.

  Example:

  (i) $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\cdots$ has the sum 1. That is $\sum_{n=1}^{\infty}\frac{1}{2^n}=1$.

  (ii) $1+2+3+4+\cdots$ has no sum.

  (iii) $1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots$. The harmonic series has no sum.

- To determine whether or not $\sum_{n=1}^{\infty} a_n$ has sum,

  (1) get the **formula for the sum for the first *k* terms**, $s_k = a_1 + a_2 + a_3 + \ldots + a_k$ (s_k is called the ***k-th partial sum*** and $\{s_k\}_{k=1}^{\infty}$ is called the ***sequence of the k-th partial sum*.**)

  (2) determine $\lim_{k\to\infty} s_k$. If $\lim_{k\to\infty} s_k = S$ exists, then $\sum_{n=1}^{\infty} a_n = \lim_{k\to\infty} s_k = S$; If $\lim_{k\to\infty} s_k$ does not exist, then $\sum_{n=1}^{\infty} a_n$ has no sum.

Definition 1:
Given a sequence of numbers $\{a_n\}$, an expression of the form

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots$$

is an **infinite series**. The number a_n is the ***n*th term** of the series. The sequence $\{s_n\}$ defined by

$$s_1 = a_1, \quad s_2 = a_1 + a_2, \quad s_3 = a_1 + a_2 + a_3,$$
$$\vdots$$
$$s_n = a_1 + a_2 + a_3 + \cdots + a_n = \textstyle\sum_{k=1}^{n} a_k,$$
$$\vdots$$

is the **sequence of partial sums** of the series, the number s_n being the ***n*th partial sum**.
If the sequence of partial sums converges to a limit L, we say that the series **converges** and that its **sum** is L. In this case, we also write

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots = \sum_{n=1}^{\infty} a_n = L.$$

If the sequence of partial sums of the series does not converge, we say that the series **diverges**.

When we begin to study a given series $a_1 + a_2 + a_3 + \cdots + a_n + \cdots$ we might not know whether it converges or diverges. In either case, it is convenient to use sigma notation to write the series as

$\sum_{n=1}^{\infty} a_n$, $\sum_{k=1}^{\infty} a_k$ or $\sum a_n$ A useful shorthand when summation from 1 to ∞ is understood.

Example 1:
Use the k-th partial sum of each series to determine whether or not the series has a sum.

(i) $1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\cdots$

(ii) $\frac{1}{2(3)}+\frac{1}{3(4)}+\frac{1}{4(5)}+\cdots+\frac{1}{(n+1)(n+2)}+\cdots$

(iii) $1+4+9+16+\ldots$

Solution:
(i)

Partial sum		Value	Suggestive expression for partial sum
First:	$s_1 = 1$	1	$2-1$
Second:	$s_2 = 1+\frac{1}{2}$	$\frac{3}{2}$	$2-\frac{1}{2}$
Third:	$s_3 = 1+\frac{1}{2}+\frac{1}{4}$	$\frac{7}{4}$	$2-\frac{1}{4}$
⋮	$S_4 =$ ⋮	$\frac{15}{8}$	$2\div\frac{1}{8}$
kth:	$s_k = 1+\frac{1}{2}+\frac{1}{4}+\cdots+\frac{1}{2^{n-1}}$	$\frac{2^n-1}{2^{n-1}}$	$2-\frac{1}{2^{k-1}}$

Sequence of the k-partial sum is $\left\{1, \frac{3}{2}, \frac{7}{4}, \cdots, 2-\frac{1}{2^{k-1}}+\cdots\right\}$.

$\lim_{k\to\infty} S_k = \lim_{k\to\infty}\left(2-\frac{1}{2^{n-1}}\right) = 2.$

$\therefore$ The series $\sum_{n=1}^{\infty}\frac{1}{2^{n-1}}$ converges and its sum is 2.

(ii) $s_k = \sum_{n=1}^{k}\frac{1}{(n+1)(n+2)} = \sum_{n=1}^{k}\left(\frac{1}{n+1}-\frac{1}{n+2}\right)$

$$=\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+\left(\frac{1}{4}-\frac{1}{5}\right)+\cdots+\left(\frac{1}{k+2}-\frac{1}{k+1}\right)+\left(\frac{1}{k+1}-\frac{1}{k+2}\right)=\frac{1}{2}-\frac{1}{k+2}$$

$\lim_{k\to\infty} s_k = \lim_{k\to\infty}\left(\frac{1}{2}-\frac{1}{k+2}\right)=\frac{1}{2}.$

$\therefore$ The series $\sum_{n=1}^{\infty}\frac{1}{(n+1)(n+2)}$ converges and its sum is $\frac{1}{2}$.

(iii) $s_k = \sum_{n=1}^{k} n^2 = 1+4+9+16+\cdots+k^2 > k^2$ for $k>1$.

Since $\lim_{k\to\infty} s_k \geq \lim_{k\to\infty} k^2 = \infty$.

Then $\lim_{k\to\infty} s_k = \infty$, the limit does not exist.

$\therefore$ The series $\sum_{n=1}^{\infty} n^2$ diverges and has no sum.

Remark:

1. To determine whether or not a **sequence** $\{a_n\}_{n=1}^{\infty}$ converges, we determine $\lim_{n\to\infty} a_n$ exists or not.

2. To determine whether or not a **series** $\sum_{n=1}^{\infty} a_n$ converges, we determine $\lim_{n\to\infty} s_n$ exists or not.

   In this case, we may need to **obtain a formula** for the n-th partial sum $s_n = \sum_{k=1}^{n} a_k$.

Some Series

➢ Geometric Series

A series of the form $\sum_{n=1}^{\infty} ar^{n-1}$ or $\sum_{n=0}^{\infty} ar^n$.

$\sum_{n=1}^{\infty} ar^{n-1}$ converges to $\frac{a}{1-r} \Leftrightarrow |r| < 1.$

Geometric series are series of the form

$$\text{series } a + ar + ar^2 + \cdots + ar^{n-1} + \cdots = \sum_{n=1}^{\infty} ar^{n-1}$$

in which a and r are fixed real numbers and $a \neq 0$. The series can also be written as $\sum_{n=0}^{\infty} ar^n$.

The **ratio** r can be positive, as in

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots \quad \text{where} \quad r = \frac{1}{2}, a = 1$$

or negative, as in

$$1 - \frac{1}{3} + \frac{1}{9} - \frac{1}{27} + \frac{1}{81} + \cdots \quad \text{where} \quad r = -\frac{1}{3}, a = 1$$

If $r = 1$, the nth partial sum of the geometric series is

$$a + a(1) + a(1) + \cdots + a(1) = an$$

and the series diverges because $\lim_{n\to\infty} S_n = \pm\infty$, depending on the sign of a.

If $r = -1$, the series diverges because the nth partial sums alternate between a and 0 and never approach a single limit.

If r ≠ 1, we can determine the convergence or divergence of the series in the following way:

s_n	=	$a + ar + ar^2 + \cdots + ar^{n-1}$	Write the nth partial sum.
rs_n	=	$ar + ar^2 + \cdots + ar^{n-1} + ar^n$	Multiply s_n by r.
$s_n - rs_n$	=	$a - ar^n$	Subtract rs_n from s_n. Most of the terms on the right cancel.
$s_n(1 - r)$	=	$a(1 - r^n)$	Factor.
s_n	=	$\frac{a(1 - r^n)}{1 - r}, (r \neq 1).$	We can solve for s_n if $r \neq 1$.

If $r < 1$, then $rn \to 0$ as $n \to \infty$ (as in Section 9.1), so $sn \to a/(1 - r)$ in this case.
On the other hand, if $r > 1$, then $rn \to \infty$ and the series diverges.

Geometric Series

If $|r| < 1$, the geometric series $a + ar + ar^2 + \cdots + ar^{n-1} + \cdots$ converges to $a/(1 - r)$:

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}, \quad |r| < 1.$$

If $|r| \geq 1$, the series diverges.

The formula $a/(1 - r)$ for the sum of a geometric series applies *only* when the summation index begins with $n = 1$ in the expression $\sum_{n=1}^{\infty} ar^{n-1}$ (or with the index $n = 0$ if we write the series as $\sum_{n=0}^{\infty} ar^n$).

Example 2:

(1) The geometric series with a = 1/9 and r =1/3 is $\frac{1}{9} + \frac{1}{27} + \frac{1}{81} + \frac{1}{16} + \cdots = \sum_{n=1}^{\infty} \frac{1}{9}\left(\frac{1}{3}\right)^{n-1}$.

Since r = 1/3 < 1, the series converges to $\frac{\frac{1}{9}}{1-\frac{1}{3}} = \frac{1}{6}$.

(2) The series. $\sum_{n=0}^{\infty} \frac{(-1)^n 5}{4^n} = 5 - \frac{5}{4} + \frac{5}{16} - \frac{5}{64} + \cdots$.

is a geometric series with a = 5 and r = −1/4. Since r = ¼ < 1, the series converges to $\frac{5}{1+\frac{1}{4}} = 4$.

(3) $\sum_{n=1}^{\infty} (-0.5)^{n-1}$ is a geometric series with $|r| = |-0.5| = 0.5 < 1$.

∴ The series converges to $\frac{1}{1-(-0.5)} = \frac{2}{3}$.

(4) $\sum_{n=1}^{\infty} e^n$ is a geometric series with $|r| = e > 1$. ∴ The series diverges.

➢ **Telescoping Series**

A series of which the terms may be expressed in the form $a_n = b_n - b_{n+1}$, and allowing one to obtain the sum through cancellation.

Example 3: $\frac{1}{2} + \frac{1}{2.3} + \frac{1}{3.4} + \frac{1}{4.5} + \cdots + \frac{1}{n(n+1)} + \cdots = \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1}\right)$.

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1},$$

so

k-th partial sum $= s_k = \sum_{n=1}^{k} \frac{1}{n(n+1)} = \sum_{n=1}^{k} \left(\frac{1}{n} - \frac{1}{n+1}\right)$

and

$$s_k = \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{k} - \frac{1}{k+1}\right).$$

(n=1, n=2, n=3, n=k)

Removing parentheses and canceling adjacent terms of opposite sign collapses the sum to

$$s_k = 1 - \frac{1}{k+1}.$$

We now see that $s_k \to 1$ as $k \to \infty$. The series converges, and its sum is 1:

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1. \quad \blacksquare$$

Example 4: Find $\sum_{n=1}^{\infty} \frac{2}{4n^2-1}$.

Solution: $s_k = \sum_{n=1}^{k} \frac{2}{4n^2-1} = \sum_{n=1}^{k} \frac{2}{(2n-1)(2n+1)}$.

Let $\frac{2}{(2n-1)(2n+1)} \equiv \frac{A}{2n-1} + \frac{B}{2n+1}$. Then $2 \equiv A(2n+1) + B(2n-1)$.

When $n = ½$, $A = 1$; When $n = -½$, $B = -1$.

$$\therefore \quad s_k = \sum_{n=1}^{k} \left(\frac{1}{2n-1} - \frac{1}{2n+1}\right) = \left(1 - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{7}\right) + \cdots + \left(\frac{1}{2k-1} - \frac{1}{2k+1}\right) = 1 - \frac{1}{2k+1}.$$

$$\lim_{k\to\infty} s_k = \lim_{k\to\infty}\left(1 - \frac{1}{2k-1}\right) = 1. \quad \therefore \text{ The series } \sum_{n=1}^{\infty} \frac{2}{4n^2-1} = \lim_{k\to\infty} s_k = 1.$$

- **Harmonic Series (Section 9.3)**

A series of the form $\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n} + \ldots$.

The harmonic series is a divergent series.

- ***p*-series (Hyperharmonic Series) (Section 9.3)**

A series of the form $\sum_{n=1}^{\infty} \frac{1}{n^p} = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \ldots + \frac{1}{n^p} + \ldots$, where p is a real constant.

$\sum_{n=1}^{\infty} \frac{1}{n^p}$ **converges if $p > 1$, diverges otherwise.**

Example 5: $\sum_{n=1}^{\infty} n^{-3}$ is a p-series with $p = 3 > 1$. ∴ The series converges.

$\sum_{n=1}^{\infty} \sqrt{n}$ is a p-series with $p = -1/2 < 1$. ∴ The series diverges.

The *n*-th Term Test for Divergence

Theorem 1: $\sum_{n=1}^{\infty} a_n$ converges $\Rightarrow$ $\lim_{n\to\infty} a_n = 0$.

Proof: $\sum_{n=1}^{\infty} a_n$ converges $\Rightarrow$ Its sum $= S = \lim_{n\to\infty} s_n \Rightarrow \lim_{n\to\infty} s_{n-1} = S$.

$$\lim_{n\to\infty} s_n - \lim_{n\to\infty} s_{n-1} = 0 \Rightarrow \lim_{n\to\infty}(s_n - s_{n-1}) = 0$$

$$\Rightarrow \lim_{n\to\infty}[(a_1 + a_2 + a_3 + \cdots + a_n) - (a_1 + a_2 + a_3 + \cdots + a_{n-1})] = 0$$

$$\Rightarrow \lim_{n\to\infty} a_n = 0$$

The *n*th-term Test for Divergence (Contrapositive of Theorem 1)

$\lim_{n\to\infty} a_n \neq 0$ **or** $\lim_{n\to\infty} a_n$ **does not exist** $\Rightarrow$ $\sum_{n=1}^{\infty} a_n$ **diverges**

REMARK: When $\lim_{n\to\infty} a_n = 0$, $\sum_{n=1}^{\infty} a_n$ may or may not converge.

Example: $\lim_{n\to\infty} \frac{1}{n^2} = 0$ and $\lim_{n\to\infty} \frac{1}{n} = 0$. But $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges and $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

❖ **DO NOT APPLY the *n*-th term test for divergence when $\lim_{n\to\infty} a_n = 0$.**

Other method is needed to determine whether $\sum_{n=1}^{\infty} a_n$ is convergent or divergent.

Example 6: Applying the *n*-th term test for divergence

(a) $\lim_{n\to\infty} n^2 = \infty$, limit does not exist. By the *n*-th term test for divergence, $\sum_{n=1}^{\infty} n^2$ diverges.

(b) $\lim_{n\to\infty} \frac{n+1}{n} = \lim_{n\to\infty}\left(1 + \frac{1}{n}\right) = 1 \neq 0$. By the *n*-th term test for divergence, $\sum_{n=1}^{\infty} \frac{n+1}{n}$ diverges.

(c) As $n \to \infty$, $(-1)^{n+1}$ alternates between 1 and -1. Therefore, $\lim_{n\to\infty}(-1)^{n+1}$ does not exist.

By the *n*-th term test for divergence, $\sum_{n=1}^{\infty}(-1)^{n+1}$ diverges.

(d) $\lim_{n\to\infty} \frac{-n}{2n+5} = \lim_{n\to\infty} \frac{-1}{2} = -\frac{1}{2} \neq 0$. By the *n*-th term test for divergence, $\sum_{n=1}^{\infty} \frac{-n}{2n+5}$ diverges.

(e) $\lim_{n\to\infty} \frac{1}{n^2+1} = 0$, further investigation is needed for $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$.

Example 7: (1) Does $\left\{\frac{2n}{n+1}\right\}_{n=1}^{\infty}$ converge? (2) Does $\sum_{n=1}^{\infty}\frac{2n}{n+1}$ converge?

Solution:

(1) $\lim_{n\to\infty}\frac{2n}{n+1}=\lim_{n\to\infty}\frac{2}{1+(1/n)}=2$. $\therefore$ The sequence $\left\{\frac{2n}{n+1}\right\}_{n=1}^{\infty}$ converges to 2.

(2) From (1), $\lim_{n\to\infty}\frac{2n}{n+1}\neq 0$. By the n-th term test for divergence, $\sum_{n=1}^{\infty}\frac{2n}{n+1}$ diverges and has no sum.

Example 8:
The series

$$1+\underbrace{\frac{1}{2}+\frac{1}{2}}_{2\text{ terms}}+\underbrace{\frac{1}{4}+\frac{1}{4}+\frac{1}{4}+\frac{1}{4}}_{4\text{ terms}}+\cdots+\underbrace{\frac{1}{2^n}+\frac{1}{2^n}+\cdots+\frac{1}{2^n}}_{2^n\text{ terms}}+\cdots$$

diverges because the terms can be grouped into infinitely many clusters each of which adds to 1, so the partial sums increase without bound. However, the terms of the series form a sequence that converges to 0. Example 1 of Section 9.3 shows that the harmonic series $\sum 1/n$ also behaves in this manner.

Theorem 2: $\sum_{n=1}^{\infty}a_n$ and $\sum_{n=1}^{\infty}b_n$ both converge $\Rightarrow \sum_{n=1}^{\infty}(a_n\pm b_n)$ and $\sum_{n=1}^{\infty}ka_n$, $k\in\mathbb{R}$, converge.

Remark: 1. $\sum_{n=1}^{\infty}a_n$ diverges $\Rightarrow \sum_{n=1}^{\infty}ka_n$ converges for all real $k\neq 0$.

2. $\sum_{n=1}^{\infty}a_n$ converges and $\sum_{n=1}^{\infty}b_n$ diverges $\Rightarrow \sum_{n=1}^{\infty}(a_n\pm b_n)$ diverges.

3. When $\sum_{n=1}^{\infty}a_n$ and $\sum_{n=1}^{\infty}b_n$ both diverges, $\sum_{n=1}^{\infty}(a_n\pm b_n)$ may or may not diverge.

Example:

(1) $\sum_{n=1}^{\infty}1$ and $\sum_{n=1}^{\infty}(-1)$ both diverge but $\sum_{n=1}^{\infty}[1+(-1)]=\sum_{n=1}^{\infty}0=0$ converges.

(2) $\sum_{n=1}^{\infty}1$ and $\sum_{n=1}^{\infty}n$ both diverge and $\sum_{n=1}^{\infty}(1+n)$ also diverges.

Example 9: Find the sum of the following series.

$$\sum_{n=1}^{\infty} \frac{3^{n-1}-1}{6^{n-1}} = \sum_{n=1}^{\infty} \left(\frac{1}{2^{n-1}} - \frac{1}{6^{n-1}} \right)$$

$$= \sum_{n=1}^{\infty} \frac{1}{2^{n-1}} - \sum_{n=1}^{\infty} \frac{1}{6^{n-1}} \quad \text{Difference Rule}$$

$$= \frac{1}{1-(1/2)} - \frac{1}{1-(1/6)} \quad \text{Geometric series with } a = 1 \text{ and } r = 1/2, 1/6$$

Both series converge since $|1/2| < 1$ and $|1/6| < 1$.

$$= 2 - \frac{6}{5} = \frac{4}{5}$$

$$\sum_{n=0}^{\infty} \frac{4}{2^n} = 4 \sum_{n=0}^{\infty} \frac{1}{2^n} \quad \text{Constant Multiple Rule}$$

$$= 4\left(\frac{1}{1-(1/2)} \right) \quad \text{Geometric series with } a = 1 \text{ and } r = 1/2$$

Converges since $|r| = 1/2 < 1$.

$$= 8$$

Example 10:
Determine whether the following series is convergent or divergent.

(1) $\displaystyle\sum_{n=1}^{\infty} \frac{\pi^{n-1}}{e^{n+1}}$ (2) $\displaystyle\sum_{n=1}^{\infty} \left(\frac{2}{n^3} - \frac{1}{3^n} \right)$ (3) $\displaystyle\sum_{n=1}^{\infty} \cos n\pi$

Solution:

(1) $\displaystyle\sum_{n=1}^{\infty} \frac{\pi^{n-1}}{e^{n+1}} = \sum_{n=1}^{\infty} \frac{1}{e^2} \left(\frac{\pi}{e} \right)^{n-1}$ is a geometric series with $a = \frac{1}{e^2}$, $|r| = \frac{\pi}{e} > 1$

$\therefore$ The series diverges.

(2) $\displaystyle\sum_{n=1}^{\infty} \left(\frac{2}{n^3} - \frac{1}{3^n} \right) = \sum_{n=1}^{\infty} \frac{2}{n^3} - \sum_{n=1}^{\infty} \frac{1}{3^n}.$

$\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^3} = 2\sum_{n=1}^{\infty} \frac{1}{n^3}$ is a multiple of a convergent p-series with $p = 3 > 1$, and

$\displaystyle\sum_{n=1}^{\infty} \frac{1}{3^n}$ is a convergent geometric series with $|r| = \frac{1}{3} < 1.$

$\therefore$ The series $\displaystyle\sum_{n=1}^{\infty} \left(\frac{1}{n^3} - \frac{1}{3^n} \right)$ converges.

(3) $\displaystyle\sum_{n=1}^{\infty} \cos n\pi = \sum_{n=1}^{\infty} (-1)^n.$

As $n \to \infty$, $(-1)^n$ alternates between 1 and -1. $\therefore$ $\displaystyle\lim_{n\to\infty} \cos n\pi = \lim_{n\to\infty} (-1)^n$ does not exist.

By the n-th term test for divergence, $\displaystyle\sum_{n=1}^{\infty} \cos n\pi$ diverges.

Adding or Deleting Terms

We can add a finite number of terms to a series or delete a finite number of terms without altering the series' convergence or divergence, although in the case of convergence this will usually change the sum. If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=k}^{\infty} a_n$ converges for any $k > 1$ and

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \cdots + a_{k-1} + \sum_{n=k}^{\infty} a_n.$$

Conversely, if $\sum_{n=k}^{\infty} a_n$ converges for any $k > 1$, then $\sum_{n=1}^{\infty} a_n$ converges. Thus,

$$\sum_{n=1}^{\infty} \frac{1}{5^n} = \frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \sum_{n=4}^{\infty} \frac{1}{5^n}$$

and

$$\sum_{n=4}^{\infty} \frac{1}{5^n} = \left(\sum_{n=1}^{\infty} \frac{1}{5^n}\right) - \frac{1}{5} - \frac{1}{25} - \frac{1}{125}.$$

Reindexing

As long as we preserve the order of its terms, we can reindex any series without altering its convergence. To raise the starting value of the index h units, replace the n in the formula for a_n by $n - h$:

replace n by n-h

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1+h}^{\infty} a_{n-h} = a_1 + a_2 + a_3 + \cdots.$$

index change according by n-h=1 ⇒ n=1+h

To lower the starting value of the index h units, replace the n in the formula for a_n by $n + h$:

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1-h}^{\infty} a_{n+h} = a_1 + a_2 + a_3 + \cdots.$$

We saw this reindexing in starting a geometric series with the index $n = 0$ instead of the index $n = 1$, but we can use any other starting index value as well. We usually give preference to indexings that lead to simple expressions.

Example 11: We can write the geometric series

$$\sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = 1 + \frac{1}{2} + \frac{1}{4} + \cdots$$

as

$$\sum_{n=0}^{\infty} \frac{1}{2^n}, \qquad \sum_{n=5}^{\infty} \frac{1}{2^{n-5}}, \qquad \text{or even} \qquad \sum_{n=-4}^{\infty} \frac{1}{2^{n+4}}.$$

The partial sums remain the same no matter what indexing we choose. ■